

ROWAN BRAD QUNI-GUDZINAS

Research & Technology Leader — Quantum Computing · Artificial Intelligence · National-Scale Systems

rowan.quni@outlook.com | ORCID: [0009-0002-4317-5604](https://orcid.org/0009-0002-4317-5604) | [ResearchGate](#)

PROFESSIONAL SUMMARY

I design architectures, build products, and publish research at the intersection of quantum computing, artificial intelligence, and large-scale systems. Over 20 years, I've: delivered national-scale platforms (AARP Livability Index); founded an AI nonprofit featured in national media (Empowering Change); and published the quantitative case that the quantum computing industry's current trajectory is a thermodynamic dead end. I bring 300+ publications, a patent portfolio, and a systematic method for cross-domain innovation. I'm seeking a VP, Director, or Research Partner role where I can set technical vision, build teams, and turn breakthrough ideas into market reality.

IMPACT HIGHLIGHTS

Demonstrated quantitatively that the quantum industry's trajectory is a thermodynamic dead end—and published a viable alternative.

Dilution refrigerators at millikelvin temperatures provide ~50 μ W of cooling. Commercial cryocoolers at 4 Kelvin provide ~1 W—a **20,000 $\times$ difference**. Published a complete roadmap for 4-Kelvin topological quantum processing using commercially available hardware and twisted high-temperature superconductors. Patents developed.

Designed quantum hardware where fault tolerance is a property of the geometry, not a software protocol.

Ultrametric Quantum Computing (UQC) uses a discrete hierarchical geometry where the mathematics itself prevents small errors from accumulating—in contrast to standard architectures that can require on the order of **1,000 physical qubits per logical qubit** for software-level error correction. UQC embeds suppression in the hardware, potentially eliminating this overhead. Published across 50+ documents. Patents developed.

Led the product that scores every neighborhood in America.

Product Manager for the AARP Livability Index: integrated **50+ data sources across 7 domains** to score every U.S. neighborhood. Full product lifecycle ownership across

multiple releases. Platform and methodology cited in **20+ academic and policy studies**. Used nationwide to guide municipal planning, grant allocation, and age-friendly policy.

Founded an AI nonprofit. Built the product. Made national news.

Took Empowering Change from incorporation through public launch. Built an LLM-powered legal navigation platform with three integrated modules for self-represented litigants. Featured in national media for pioneering AI-driven legal democratization. Validated product design firsthand by serving as a pro se litigant in San Francisco tenant rights cases.

Managed \$1.5M in federal R&D contracts with full procurement authority.

Certified Contracting Officer's Representative at U.S. Department of Transportation. Full oversight of technical performance, budgets, milestones, and contractor deliverables. Led key components of a national forecasting model that informs federal infrastructure investment. Clearance-eligible.

300+ publications, a patent portfolio, COR certification, and twenty years of sustained output across government, consulting, nonprofit, and independent research.

Open-access body of work spanning quantum architecture, AI/ML systems, and signal processing—freely accessible on Zenodo and ResearchGate. Patent portfolio covering quantum computing architectures, neural network designs, and foundational computing paradigms.

PROFESSIONAL EXPERIENCE

Principal Investigator & Founder

QWAV / QNFO—Independent Research | 2024–Present

Research Output: Published 300+ scholarly works spanning quantum computing architecture, AI/ML systems, signal processing, and cross-domain methodology—all open-access on Zenodo, ResearchGate, and SSRN. Self-directed program spanning quantum architecture, AI/ML, and signal processing.

Quantum Computing: Published the quantitative case against dilution-refrigerated quantum architectures: $\sim 50 \mu\text{W}$ of cooling at millikelvin versus $\sim 1 \text{ W}$ at 4 Kelvin—a $20,000\times$ gap that fundamentally limits scalability. Published a complete 4-Kelvin topological processing roadmap using 45° twisted Bi-2212 superconductors (predicted thermal stability margin $\Gamma \approx 80$). Designed Ultrametric Quantum Computing (UQC), embedding error suppression in the hardware geometry—potentially eliminating the 1,000:1 qubit overhead of conventional error correction. Proposed Quantum Resonance Computing (QRC), a field-based alternative paradigm. Published across 50+ documents. Patents developed.

Artificial Intelligence: Proposed Prime-Attentive Neural Networks (PANNs), a neural architecture incorporating number-theoretic organizational principles. Published the Alpha Pi Project, a 7-chapter monograph adapting cardiac signal processing techniques to quantum state readout. Identified the fundamental signal-processing ceiling imposed by continuous-state architectures—a finding that directly motivated the UQC design.

Intellectual Property & Credentials: Developed patent portfolio covering quantum architectures (UQC, QRC, 4K topological), neural networks (PANNs), and foundational computing paradigms. Applied for the QISTgrad PhD Fellowship in Quantum Science and Technology (January 2026).

Executive Director & Founder

Empowering Change—501(c)(3) Nonprofit | San Francisco, CA | 2023–2024

Founded the organization and led it from incorporation through public launch. Built an LLM-powered legal navigation platform with three integrated modules: document drafting, procedural guidance, and legal jargon translation for self-represented litigants.

Featured in national media for pioneering AI-driven legal democratization.

Validated product design firsthand by serving as a pro se litigant in San Francisco tenant rights cases, using direct experience to inform platform requirements.

Product Manager & Senior Methods Advisor

AARP Public Policy Institute | Washington, DC | 2016–2021

Led the AARP Livability Index—a national platform integrating 50+ distinct data sources across 7 domains to score every U.S. neighborhood. Owned the full product lifecycle from strategy through multiple public releases.

Generated measurable policy influence. Platform and methodology cited in 20+ academic and policy studies. Used nationwide to inform municipal planning, community development grant allocation, and age-friendly policy.

Co-authored the Livability Index 2018 report and multiple AARP policy briefs on housing affordability, transportation access, and age-friendly community development.

Managed complex multi-stakeholder relationships spanning federal agencies, state and local governments, academic partners, advocacy organizations, and internal AARP divisions.

Senior Associate, Data Analytics

Deloitte Consulting LLP | Washington, DC | 2015–2016

Led data analytics and machine learning engagements for federal and public-sector clients. Deployed predictive models and data integration strategies addressing

complex organizational challenges.

Operated within a top-tier consulting environment, combining technical delivery with client management, executive communication, and business development.

Data Analyst & Research Manager

U.S. Department of Transportation—Federal Highway Administration (FHWA) |
Washington, DC | 2011–2015

Managed a federal research portfolio exceeding \$1.5M as certified Contracting Officer's Representative. Full authority for technical oversight, budgeting, milestone tracking, and contractor deliverable acceptance.

Led key technical components of the national Tour-Based Model System for long-distance passenger travel forecasting. Model output directly informs federal infrastructure investment decisions.

Coordinated interagency modeling standards across federal, state, and academic partners. Published peer-reviewed research on multimodal travel behavior and national transportation forecasting.

Earlier Professional Roles

Role	Organization	Period
Platform Data Product Manager	iManage	2022–2023
Senior Product Manager	Epsilon (Publicis Groupe)	2021–2022
Senior Data Scientist	The Advisory Board Company	2016
Consultant	IBI Group	2008–2011
Systems Planner	Delaware Valley Regional Planning Commission	2005–2008
GIS Analyst	Cross County Connection TMA	2004–2005

TECHNICAL EXPERTISE

Area	Capabilities
Quantum Computing	Architecture design (UQC, QRC, 4K topological), cryogenic systems analysis, topological quantum computing, hardware-level error suppression, technology roadmapping, quantum-classical interfaces
AI & Machine Learning	Neural architecture design (PANNs), LLM deployment and productization, applied ML/NLP, predictive modeling, signal processing, AI product development
Product & Platform Leadership	Full product lifecycle, product strategy and roadmap, multi-stakeholder management, go-to-market planning, cross-functional team building, user-centered design
Data & Analytics Architecture	Large-scale data integration (50+ heterogeneous sources), national-scale statistical modeling, geospatial analytics (GIS), real-time pipeline design, federal data standards
Research & Innovation Management	Independent research program design, patent strategy, open science dissemination, technology commercialization, federal grant and contract management (COR)
Cross-Domain Methodology	Structural pattern recognition across disparate systems, complex systems modeling, multi-scale architecture, emergence and feedback analysis, formal isomorphism identification

PUBLICATIONS & INTELLECTUAL PROPERTY

300+ scholarly works published open-access on Zenodo, ResearchGate, and SSRN—freely accessible, no paywalls. Works span preprints, technical reports, and books across quantum architecture, AI/ML, signal processing, and cross-domain methodology.

Patent portfolio covering quantum computing architectures (UQC, QRC, 4-Kelvin topological processing), neural network designs (Prime-Attentive Neural Networks), and foundational computing paradigms.

Representative works:

Orchestrating the Quantum Future—4-Kelvin quantum computing roadmap (v1.0–v1.1)

Ultrametric Quantum Computation—UQC architecture series (v0.1–v1.0, 50+ documents)

Quantum Resonance Computing—Field-based quantum paradigm proposal

The Thermodynamic Imperative—Quantitative analysis of quantum scaling limits

Alpha Pi Project—7-chapter monograph: cardiac signal processing → quantum architecture

The Ultrametric Paradigm—Synthesis across multiple versions (v0.1–v0.9)

A General Theory of Process—Cross-domain isomorphism framework

The Information Spectrum—Book-length treatment of information theory

EDUCATION

M.S., City and Regional Studies—Rutgers University, Edward J. Bloustein School of Planning and Public Policy (2008)

B.A., Urban Studies and Community Planning—Rutgers University–Camden (2004)

CERTIFICATIONS & PROFESSIONAL IDENTIFIERS

Project Management Professional (PMP)—Project Management Institute (2016)

Contracting Officer's Representative (COR)—Defense Acquisition University (2011) · *Clearance-eligible*

American Institute of Certified Planners (AICP)—American Planning Association (2009)

ORCID: 0009-0002-4317-5604 · **ISNI:** 0000 0005 2645 6062

QISTgrad PhD Fellowship—Application submitted (January 2026)

TARGET ROLES

Sector	Representative Roles	Why This Profile Fits
Quantum Computing	VP/Director of Quantum Research, Architecture, Hardware, or Engineering; Chief Quantum Scientist; Distinguished Engineer	Published architectures (UQC, QRC, 4K topological). Quantitative scaling analysis. 50+ technical documents. Patent portfolio
AI & Machine Learning	VP/Director of AI Research, Applied AI, or ML Engineering; Chief AI Scientist	PANNs architecture. Launched LLM-based platform. Data science leadership at national scale. 300+ publications
Big Tech & Platforms	Director of Research, VP Emerging Technology, Distinguished Scientist, Director of Advanced Technology	Cross-domain expertise (quantum + AI + systems). Product leadership at national scale. Patent portfolio
Defense & National Security	Director of Quantum Research, Chief Scientist, VP Advanced Technology, Senior Technical Fellow or Advisor	COR certification (clearance-eligible). Federal agency experience. Quantum architectures. Complex systems modeling
Government Labs & FFRDCs	Director of Quantum Research, Chief Scientist, Program Director, Senior Technical Advisor	COR certification. Federal program management. Quantum computing R&D. Publication record
Venture Capital	Research Partner, Technical Fellow, Entrepreneur in Residence, Director of Deep Tech Research	Technical evaluation (quantum + AI). 300+ publications. Founded two organizations. Product leadership
Consulting	Director/VP of Quantum Practice, VP of AI & Advanced Analytics, Senior Expert/Fellow, Practice Lead for Emerging Technology	Prior Deloitte experience. Client communication. 300+ publications. Cross-domain methodology
Startups & Scale-ups	CTO, Co-Founder/CEO, VP of Engineering, Head of Research, Chief Product Officer	Founded two organizations. Product leadership (AARP, Empowering Change). Patent strategy. Technical vision
Financial Services	VP of Quantum or Advanced Computing, Director of Advanced Analytics, Head of Computational Research	Quantum architectures for optimization and risk. Data science at scale. Systems modeling

Sector	Representative Roles	Why This Profile Fits
Policy & Think Tanks	Senior Fellow, Director of Technology or Innovation Policy, Research Director	300+ publications. Federal policy experience (AARP, FHWA). Cross-domain expertise

NOTES

Name History: Formerly known professionally as Brad Gudzinis. All research since 2024 published under Rowan Brad Quni-Gudzinis.

Location: Open to remote, hybrid, or on-site. Willing to relocate.